Popularity, Leakage, or Structure? Auditing Biomedical Link Prediction on `ogbl-biokg`

Topic C3 · Biomedical link prediction on `ogbl-biokg` · Team 7: Kamil Gainutdinov, Artem Salakhutdinov, Marat Akhmetov · October 2026

1 · Task

We study knowledge-graph completion on a biomedical KG: given $(h, r, ?)$ or $(?, r, t)$, rank the true entity against 500 sampled same-type negatives, following the official OGB protocol. Uniform same-type negatives are "too easy" (DKR Lecture 3, Pitfall 5 [5]), and the headline numbers show it: a scorer that never looks at the head already reaches test MRR 0.676 (learned tail-and-relation model) or 0.339 (raw count prior), against 0.0136 for chance and 0.810 for ComplEx [1,6, Table 3]. We ask: *how much of ComplEx-level MRR is (i) answer popularity, (ii) lookup of the reverse triple, and (iii) genuine head-dependent structure — and on which relations does (iii) survive?* Success measure: a per-relation × frequency-stratified evaluation along a ladder of baselines (chance → count prior → learned head-blind scorer → reverse lookup → TransE/ComplEx) with paired confidence intervals.

2 · Dataset

`ogbl-biokg` from the Open Graph Benchmark [1]: 93,773 entities in 5 types (function 45,085; protein 17,499; disease 10,687; drug 10,533; side effect 9,969), 51 directed relation types, 5,088,434 triples, official random split 4,762,678 / 162,886 / 162,870 train/valid/test. For every valid/test triple OGB gives 500 head and 500 tail negatives of the correct type, filtered against the KG; each side is ranked against its own 500 separately. No node features. Relations between entities of the same type (drug–drug, protein–protein, function–function) are, per OGB, always symmetric and stored in both directions [1] — we verify this — while the split is random, so a test triple's reverse can sit in train (≈94% under independent edge splits). Entity frequency is highly skewed [7]. Free Colab, target $\leq$ 8 GPU-hours.

3 · Method

All learned models share one pipeline — official splits, the same type-constrained negative-sampling loss, 400 real parameters per entity, the same number of steps — so differences come from the scorers. (i) *Baseline ladder:* a random scorer (continuous scores); the *count prior* $s(h,r,t) = \#\{\text{train triples with } t \text{ as a tail of } r\}$ (head queries: symmetric), i.e. EntOccur [6], the null model of [8]; a learned *head-blind* scorer $\langle \mathbf{r}, \mathbf{e}_t \rangle + b_t$ in the spirit of AutoWeird [6]; and an *inverse-lookup* scorer $[(t,r,h) \in \text{train}] + \varepsilon \cdot$prior. (ii) *KG embeddings*, implemented by us in PyTorch in the style of Lab 4 and cross-checked against `torch_geometric.nn.kge` on fixed batches: TransE [2] (d=400) and ComplEx [3] (complex rank 200) (parameter-matched capacity control), one shared trainer, three seeds each. Our contribution is an audit, not a model: a stratified harness; a frequency-matched evaluation, i.e. the "highly correlated negatives" protocol that [6] proposed but left as future work; per-relation attribution along the ladder (symmetry, 1-to-N, tail concentration; cf. [4]); and an inverse-triple leakage audit (explicit reverse lookup only, not indirect paths).

4 · Evaluation plan

The protocol is fixed before the final runs. *Metrics:* MRR and Hits@1/3/10; head and tail queries separately and pooled; per-query values come from the official evaluator (`ogb` 1.3.6, `mrr_list`), which ranks ties at the average rank, $1 + \#\{\text{strictly higher}\} + \frac{1}{2}\#\{\text{equal}\}$. *Strata:* (A) answer frequency = train count of the answer in the same role for the same relation; per relation and direction, a zero-count bin plus tertiles over the rest, edges computed on validation and frozen before test is touched, tied counts kept in one bin; (B) for same-type relations, whether the reverse triple is in train. Cells with <30 test queries are pooled into their relation group (head-type/tail-type pair). *Frequency-matched control:* each test query is re-ranked against 500 negatives drawn from the 1,000 same-type entities closest to the answer in train-frequency rank (same relation and role; for zero-count answers the pool is all same-type zero-count entities, frequency ties broken by a fixed entity-id order), filtered against train$\cup$valid$\cup$test, drawn with 3 fixed seeds (inference only; we report the spread) and shared by all scorers; the count prior is near chance there by construction, so it is a harness check, not a finding. *Rigor:* 3 seeds, mean $\pm$ std and the three values; paired *cluster* bootstrap over undirected test edges (head and tail queries of one triple, and both directions of a symmetric edge, resampled together; seed-averaged scores, 1,000 resamples, 95% CIs). Confirmatory family (BH-FDR, $q = 0.05$): H1–H3 and the pooled ladder margins; all per-relation cells (≈400 in the grid) are exploratory, reported as effect size + CI. One ablation — ComplEx with vs. without OGB's frequency-based triple-subsampling weight. *Sanity gates:* continuous random scorer within 0.0136 ± 0.0005 (analytic $H_{501}/501$ over 501 candidates); no direct train/test overlap; same-type triples have their reverse somewhere in the KG; our ranks equal the official evaluator on a fixed batch; the count prior reproduces EntOccur's 0.3387 [6, Table 3] inside our pessimistic–optimistic tie bracket (predating ogb 1.3.4, PyPI 2022-08-20, which replaced the old argsort rule — ties implementation-defined — with mid-rank; any tie rule keeps each rank inside $[\text{pess}, \text{opt}]$); for reference, published TransE/ComplEx (0.745/0.810) use 188M parameters (≈2,000 per entity) [6, Table 3] — 5× our 37.5M ($93{,}773 \times 400$). *Hypotheses:* **H1** — per-relation count-prior MRR tracks tail concentration (Spearman $\rho \geq 0.5$ with the Gini of the tail distribution, across relations). **H2** — on same-type relations, the lookup rung scores far higher on queries whose reverse is in train, and the TransE–ComplEx gap is larger on that stratum (TransE cannot represent symmetry [4]). **H3 (main)** — head-dependence survives frequency matching: $G = \text{MRR}(\text{ComplEx}) - \text{MRR}(\text{head-blind}) > 0$ on matched negatives (CI excludes 0); we also report $G_{\text{official}} - G_{\text{matched}}$.

5 · Main risk and fallback

The experiment that could kill the idea runs first, in week 1, as three numbers: (a) the share of same-type test triples whose reverse is in train, (b) the MRR of every ladder rung under the official protocol, stratified, (c) a small ComplEx and the head-blind rung on official vs. frequency-matched negatives. If the paired drop in (c) is <0.02 in every bin and (a) is small, neither popularity nor leakage explains much of the score; the opposite outcome is also a finding — the head-blind rung already recovers most of ComplEx. If (a) is near 94%, the no-reverse stratum is small and its CIs wide, and we report that most same-type test queries are lookups. Compute fallback: smaller rank, 2 seeds, fixed step budget, Drive checkpoints. Caveats: rare entities also have less training signal, so frequency bins alone cannot separate popularity bias from data scarcity; matched negatives near popular answers are themselves popular, hence more likely true-but-missing links, so a drop mixes "popularity no longer helps" with "harder negatives". Scoring the ladder on the same negatives only partly controls for this; we claim what the metric rewards, not why.

Timeline — Oct 23–29: data, ladder, harness, week-1 test · by Nov 12: two models × 3 seeds, stratified results (midway) · Nov 13–25: CIs, matched control, ablation, leakage · Nov 26–Dec 3: report, repo, talk.

References — [1] Hu et al., Open Graph Benchmark, NeurIPS 2020. [2] Bordes et al., TransE, NeurIPS 2013. [3] Trouillon et al., ComplEx, ICML 2016. [4] Rossi et al., KG Embedding for Link Prediction: A Comparative Analysis, ACM TKDD 2021. [5] DKR Fall 2026, Lecture 3 (Pitfall 5), Lecture 4, Lab 4. [6] Yang, Zhang, Yao, AutoWeird, arXiv:2207.11673, 2022. [7] Bonner et al., Topological Imbalance in Biomedical KGs, Brief. Bioinform. 2022. [8] Douglas et al., What is Learned in Knowledge Graph Embeddings?, arXiv:2110.09978, 2021.